Chapter 5 : Data Compression.

X = random V.
$\mathcal{X}$ = sample space.
$\mathcal{D} = \{0, 1, \ldots, D-1\}$

Def A source code $C : \mathcal{X} \to \mathcal{D}^*$

ex) $D = 2$. $\mathcal{D} = \{0,1\}$. $\mathcal{D}^* = \{01, 000, 101, 111, \ldots\}$.

• Expected length of C : $E\ell(X) = \sum_{x \in \mathcal{X}} p(x)\ell(x) =: L(C)$

$\boxed{\text{Instantaneous Code (prefix)}}$

$\Downarrow$

Uniquely decodable.

$\Downarrow$

Nonsingular.

$\Downarrow$

All Code.

$\mathcal{X}$	C	
a	0 1	0 1
b	1 1	1 0
c	1 1 1	1 1 0

0 1 | 1 0 | 1 1 0 | 0 1 1

a	1 0
b	0 1
c	1 1 0
d	0 1 0
e	1 0

$\boxed{\text{Thm}}$ (Kraft inequality).

Instantaneous Code $\Longrightarrow \sum_i D^{-l_i} \le 1$.

$\Longleftarrow$ construct.

(Mcmillan inequality)

Uniquely Decodable codes : The same situation holds.

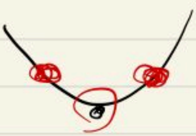
• min $E\,l(X) = L(C) = \sum_i p_i l_i$

subject to $\sum_i D^{-l_i} \le 1$, $\cancel{l_i \in \mathbb{N}}$

Indeed $l_i^* = -\log_D p_i \Rightarrow$ global minimum

$$\text{global minimum} = -\sum p_i \log_D p_i = H_D(X)$$

$(x-1.5)^2$



$x = 1.5$

$\wedge$

$p_i = D^{-n}$

D-adic

(Thm) 「Not that too far away」

$l_1^*, l_2^*, \dots, l_m^*$ optimal codeword lengths for a D-ary alphabet. $L^* = \sum p_i l_i^*$

$$H_D(X) \le L^* < H_D(X) + 1$$

∵ $l(x) = \left\lceil \log \frac{1}{p(x)} \right\rceil$ $\left(\begin{array}{l} \lceil 2.8 \rceil = 3 \\ \lceil 1.5 \rceil = 2 \end{array} \right)$

Shannon Code.

Interest : Optimal & instantaneous
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\quad \| $
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad$ prefix

Huffman Code.

ex (Wikipedia)

A_EADD_DAD_CEDED_A_BAD_BABE_A_BEADED_ABACA_BED.

① A : 11
B : 6
C : 2
D : 10
E : 17
_ : 10

⇒ ② C : 2
B : 6
E : 7
_ : 10
D : 10
A : 11.

CB = 8

0 1

C B

⇒ E = 7
CB = 8
_ = 10
D = 10
A = 11

ECB = 15

0 1

E CB

$P_i \times l_i$

A :	10
B :	1111
C :	1110
D :	01
E :	110
_ :	00

10 00 110 100101 .....

Existence of Optimal Code? $\Rightarrow$ Yes

$\Downarrow$

Uniqueness of Optimal Code $\Rightarrow$ No.

$\because$ can make many optimal codes

$\Downarrow$

How to find it?

10 → 01
01 → 10
11 → 000
11

$\because$ Well...... Huffman Code.

$\boxed{\text{Thm}}$ (Huffman Coding is Optimal)

pf) Use induction on # of symbols.

Assume $p = (p_1, p_2, \ldots, p_m)$ $p_1 \geq p_2 \geq \cdots \geq p_m$.

$\Downarrow$ condense

$p' = (p_1, p_2, \ldots, p_{m-2}, p_{m-1} + p_m)$

Ref : (Canonical Optimal Code)

1. $p_i > p_j \Rightarrow l_i \leq l_j$
2. Two longest codewords
   1. have the same lengths — 11100
   2. differ only in the last bit — 11101
   3. correspond to least likely symbols.

We have $C_m^*(p)$: Canonical Optimal Code.

$C_{m-1}^*(p')$: Optimal Code.

<u>Claim</u> Extension of Optimal code for p' is Optimal for p.

p' $\quad C_{m-1}^*(p') \quad L^*(p') \quad \rightsquigarrow \quad C_m(p)$

p'	$C_{m-1}^*(p')$		$C_m(p)$	
p_1	ω_1'	l_1'	$\omega_1 = \omega_1'$	$l_1 = l_1'$
p_2	ω_2'	l_2'	$\omega_2 = \omega_2'$	$\vdots$
$\vdots$		$\vdots$	$\vdots$	
p_{m-2}	ω_{m-2}'	l_{m-2}'	$\omega_{m-2} = \omega_{m-2}'$	$l_{m-2} = l_{m-2}'$
$p_{m-1}+p_m$	ω_{m-1}'	l_{m-1}'	$\omega_{m-1} = \omega_{m-1}'0$	$l_{m-1} = l_{m-1}'+1$
			$\omega_m = \omega_{m-1}'1$	$l_m = l_{m-1}'+1$

$$\boxed{L(p) = L^*(p') + p_{m-1} + p_m}$$

Now from $C_m^*(p)$ construct a code p' by Huffman reduction.

$$\boxed{L(p')} = \sum_{i=1}^{m-2} p_i l_i + p_{m-1}(l_{m-1}-1) + p_m(l_m-1)$$

$$\boxed{= L^*(p) - p_{m-1} - p_m}$$

$$\Rightarrow L(p) - L^*(p') = L^*(p) - L(p')$$

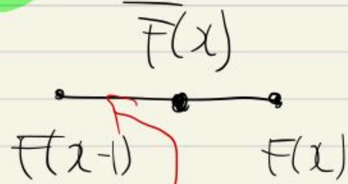
$$\underbrace{\left(L(p) - L^*(p)\right)}_{\geq 0} + \underbrace{\left(L(p') - L^*(p')\right)}_{\geq 0} = 0$$

$L(p) = L^*(p)$

Shannon-Fano-Elias Coding

$$F(x) = \sum_{i \le x} p(i) \underset{\text{modify}}{\rightsquigarrow} \bar{F}(x) = \sum_{i<x} p(i) + \frac{1}{2}p(x)$$

$$\rightsquigarrow \lfloor \bar{F}(x) \rfloor_{l(x)}$$ (truncated $\bar{F}$)



e.g. $\bar{F} = 0.\underline{11010}1001$

$l(x) = 5$

$$\lfloor \bar{F} \rfloor_{l} = 0.11010$$

$$\boxed{l(x) = \left\lceil \log \frac{1}{p(x)} \right\rceil + 1}$$

$$\Rightarrow F(x-1) < \lfloor \bar{F}(x) \rfloor_{l(x)} < F(x)$$

• Notion of prefix-free

$$\begin{cases} 0.11010|10 \\ 0.11010|11 \end{cases}$$

Given $\lfloor \bar{F}(x) \rfloor_{l(x)} = 0.z_1 z_2 \cdots z_l$

$$\Longleftrightarrow \left[0.z_1 \cdots z_l,\ 0.z_1 \cdots z_l + \frac{1}{2^l}\right)$$

prefix-free code $\Longleftrightarrow$ such intervals are disjoint.

Shannon Code. $l(x) = \left\lceil \log \frac{1}{p(x)} \right\rceil$

$\boxed{\text{Thm}}$ In Shannon Code, $l'(x)$: any uniquely decodable code length

$$\Rightarrow \Pr(l(X) \geq l'(X) + c) \leq \frac{1}{2^{c-1}}$$

i.e. "$l(X)$ is quite short".

(pf) $\Pr\left(\left\lceil \log \frac{1}{p(X)} \right\rceil \geq l'(X) + c \right)$

$$\leq \Pr\left(1 + \log \frac{1}{p(X)} \geq l'(X) + c\right)$$

$$= \Pr\left(p(X) \leq 2^{-l'(X) - c + 1}\right)$$

$$= \sum_{p(x) \leq 2^{-l'(x) - c + 1}} p(x) \leq \sum_{p(x) \leq 2^{-l'(x) - c + 1}} 2^{-l'(x) - c + 1}$$

$$\leq \sum_x 2^{-l'(x) - c + 1}$$

$$\leq 2^{-(c-1)}$$

$\square$

$\boxed{\text{Thm}}$ $p(x)$ = dyadic prob. mass function

$l(x) = \log_2 \frac{1}{p(x)}$, l' : other uniquely decodable binary code.

Then, $\Pr(l(X) < l'(X)) \geq \Pr(l(X) > l'(X))$

(pf) $\Pr(l < l') - \Pr(l > l')$

$\operatorname{sgn}(x) = \begin{cases} 1 & x>0 \\ 0 & x=0 \\ -1 & x<0 \end{cases}$

$$= \sum_{l<l'} p(x) - \sum_{l>l'} p(x)$$

$$= \sum_x p(x) \operatorname{sgn}(l(x) - l'(x)) \quad (*)$$

Note $\operatorname{sgn}(x) \leq 2^x - 1 \quad x \in \mathbb{Z}$.

$$(*) \leq \sum_x p(x)\left(2^{l(x)-l'(x)} - 1\right)$$

$$= \sum_x 2^{-l(x)}\left(2^{l(x)-l'(x)} - 1\right)$$

$$= \underbrace{\sum_x 2^{-l'(x)}}_{\leq 1} - \underbrace{\sum_x 2^{-l(x)}}_{=1} \leq 0$$

$2^{-l(x)} = p(x)$

$\square$

<u>Rmk</u>: " $=$ " $\iff \ell'(x) = \ell(x) \ \forall x \in X$.